

Calculation Walkthroughs — How Every Result-Panel Number Is Computed

One markdown file per test scenario in the parent folder. Load the scenario, calculate, then read the matching file side-by-side with the result panel: each section of the file mirrors a section of the panel and shows the arithmetic behind the numbers on screen.

All figures were verified against a live API instance built from commit `559aef2` (2026-07-22). Fuel price is **\$780/ton** unless the file says otherwise — \$ figures scale linearly with price.

The 7-step recipe (used by every walkthrough)

Step 1 — Battery cascade (Excel "Load Demands" sheet). The battery power budget is handed out over the loads in strict priority order: DpReserve → DpDemand → Mission → Propulsion → Hotel. Per row:

H = what the row wants	PEAK SHAVING rows: avg load × variation factor (propulsion 5%, hotel 2%)
	MISSION row: the full heavy-consumer max (it can start any moment)
	RESERVE row: the full requirement (class rule, kW for kW)
I = min(remaining budget, H)	what the row takes from the budget
J = I × coverage factor	what actually counts as covered
	(RESERVE 100% · DP/Mission 50% · propulsion 35% · hotel 5%)
L = H - J	what the gensets must still carry (spinning reserve)
remaining -= I	the next row sees a smaller budget

Tiles: **Peak Shaving** = $\sum J$ (peak-shaving rows) · **Spinning Reserve** = $\sum L$. Invariant: $\sum J + \sum L = \sum H$ — the sea sets the total swing; the battery only moves the split.

Step 2 — Adjusted demands. Per side: `propulsion' = propulsion_avg + L(propulsion-side rows)`, `hotel' = hotel_avg + L(hotel-side rows: hotel, mission, DP-hotel)`. Covered J never reduces demand.

Step 3 — Load distribution (per candidate combination ME-count × SG × AE-count):

SG power = min(hotel', SG capacity of ACTIVE MEs)	SG covers hotel first
AE power = min(hotel' - SG power, AE capacity)	AE takes the remainder
ME power = propulsion' + SG power	the SG is a shaft load ON the ME
load % = power / capacity of the ACTIVE units	

Step 4 — SFOC → FOC. `FOC [t/h] = P_ME × SFOC_ME(load%)/1e6 + P_AE × SFOC_AE(load%)/1e6`. SFOC is linearly interpolated from the selected engine type's curve (DB), extrapolated below the lowest point.

Step 5 — Ranking. Sort valid combos by FOC. **Optimal = first row** (this IS Integration Level 1 — it is not user-selectable). **Baseline** = last row (no battery) or **3rd-from-worst** (battery active, clamped for short lists); the radio in "Assumed Configuration" can override it.

Step 6 — Annual numbers. `FOC t/yr = t/h × mode hours (summed over modes) · CO2 = FOC × per-fuel factor (MGO/MDO 3.93267 · HFO 3.114 · LNG 2.753)` — ME and AE each use their own fuel's factor · `cost =`

$\text{FOC} \times \text{fuel price} \cdot \text{iEMS savings} = \text{baseline} - \text{optimal}$ · tier investments are fixed constants (\$110k / \$140.5k / \$210k).

Step 7 — Battery Benefit (the green badge; dual-scenario rule R3a). The code runs Level 1 twice: world A (with battery, demand = avg + Σ L) and world B (reference: budget 0, demand = avg + Σ H). $\text{Benefit} = \max(0, \text{FOC_B_optimal} - \text{FOC_A_optimal}) \times \text{hours}$, summed per mode; $\$ = \text{benefit} \times \text{fuel price}$. The baseline radio never touches this number.

Files

01–12 mirror the first-wave scenarios, 13–18 the second wave. Error scenarios (09, 17) explain the rejection math instead of panels; 18 explains the deliberate absence of the battery panel.